Education Project – Socioeconomic Factors and ACT Scores

Abstract

This project analyzes data from a small subset of schools in the United States to determine if socioeconomic factors predict the average ACT score of a school and if so, which socioeconomic factors give the best prediction. This project uses regression to fit single predictor and multiple predictor models. The single predictor models of median income and number of teachers explained 21.1% and 3.0% of the variance in average ACT score. A multiple regression model with five socioeconomic predictors explained 62.8% of the variance, with a mean absolute error (MAE) of 1.15 ACT points. A reduced model with unemployment rate, percent college, and percent lunch performed similarly. Out of all the predictor variables considered, the percent of students receiving free or reduced price lunch was the strongest predictor.

Introduction

This project analyzes school level data to determine which socioeconomic factors best predict the average ACT score of a school. To do this, the project combines three public data sets. The EdGap data set provides the average ACT score (or SAT equivalent) and five socioeconomic measures of each school: the percent of students receiving free or reduced price lunch, unemployment rate, percent of adults with a college degree, percent of children in married couple families, and median household income. Two data sets from the school directory from the NCES Common Core of Data for the 2016–2017 school year were also used. One provides the state, ZIP code, school type, and school level. The other provides the number of teachers per school for the school year, approximating the size of the school. This variable is not adjusted to number of teachers per student and is a rough approximation of school size.

The data for this analysis is a table of data stored in the file education_clean.csv. The response variable is average_act, the average ACT score of a school (about 20.3 on average). Six numeric predictors are available:

- rate_unemployment: unemployment rate in the school's area (proportion, 0–1)
- percent_college: share of the area's adults with college education (proportion)
- percent_married: share of married households (proportion)
- median_income: median household income (US dollars)
- percent_lunch: share of students receiving free or reduced-price lunch (proportion)
- teachers: number of teachers at the school (count)

Each row also carries identifiers and descriptors of the school (state, ZIP code, school type, and charter status).

The data is first explored with a correlation matrix, pair plots, and boxplots. The analysis then proceeds from simple to complex: single-predictor models, a multiple regression, a reduced model chosen by significance testing, and a standardized version for comparing effect sizes.

Theoretical Background

This project uses linear regression modeling. The model calculates a response variable, which is the average ACT score for this project, as a weighted sum of predictor variables. The model chooses the weights so that the predictions are as close as possible to the actual values. Adding a squared term, such as income squared, creates a quadratic model and generates a curve to fit the data instead of a straight line.

Model fit is measured with R^2, the variation in ACT score between the model; with the root mean squared error (RMSE), and also with mean absolute error (MAE), which describes the typical size of the prediction errors in ACT points. Each predictor is tested for significance with a t-test. Models are compared with an F-test, and a small p-value means the extra terms improve the fit. Residual plots, which graph the prediction errors against the predicted values, shows whether a linear model is appropriate. Finally, standardizing the predictors to have mean 0 and standard deviation 1 makes the coefficients comparable, because each one is then the change in ACT score for a one standard deviation change in that predictor.

Methodology

This report looks at the relationship between average ACT score and socioeconomic variables in three parts: (1) exploratory analysis of the distributions of and correlations between the variables, (2) OLS regression models with a single predictor and with multiple predictors, and (3) a comparison of the models to find a simple model that fits as well as the more complicated ones. The analysis is done in Python using pandas and NumPy for data handling, statsmodels for OLS regression and nested model tests, scikit-learn for metrics, scaling, and imputation, and matplotlib and seaborn for plots.

The data come from three files. The first, with socioeconomic data and ACT scores is the EdGap data (EdGap_data.xlsx). The other two files are from the NCES Common Core of Data school directory for 2016–2017.

The school ID is read as an identifier (type object) rather than a number so that it can be used as a join key. The first school directory is reduced to the columns for year, school ID, district ID, state, ZIP code, school type, and school level, joined to the second school directory to include number of teachers, and then left joined to the EdGap data by school ID so that every EdGap school is kept. Columns are renamed to descriptive snake case names. Impossible values are set to missing (percent lunch below 0 and average ACT score below 1).

Since this analysis is focused around ACT scores, the data set was limited to only high schools, as the ACT is typically taken in high school as a college entrance examination. Also, schools with a missing ACT score average were dropped from the dataset.

The EdGap data is limited to a select number of states, so the final merged data is limited to a select number of states, as shown in Figure 1.

Figure 1. Distribution of school data by state



Computational Results

From the exploratory data analysis, it is evident all six predictors are correlated with average ACT score, and the pair plots show relationships that are roughly linear. The predictors are also correlated with each other (Table 1), most strongly median income with percent college (0.71) and percent married (0.62). This overlap suggests that not all predictors are needed in a model. Percent lunch has the strongest correlation with average ACT score (−0.78), and the teacher count has the weakest (0.17).

Table 1. Correlation of each predictor with average ACT score and with the most strongly correlated other predictor

Predictor Variable	Correlation Average ACT	Strongest Correlation with another Predictor
percent_lunch	−0.78	−0.52 (percent_married)
median_income	0.46	0.71 (percent_college)
percent_college	0.46	0.71 (median_income)
percent_married	0.44	0.62 (median_income)
rate_unemployment	−0.43	−0.56 (percent_married)
teachers	0.17	0.31 (median_income)

Boxplots show outliers in unemployment rate, percent college, percent married, median income, and number of teachers. None of these are obviously incorrect values. They likely reflect skewed distributions, so all observations were kept.

For the linear regression, the residuals were scattered around zero, but their spread was not constant across predicted values. This suggested that the linear regression model may not fully capture the patterns in the data. Other models were generated, as shown below in Table 2.

Table 2. Fit of statistical models

Model	Predictors	R^2	MAE (ACT points)
Linear	median income	0.211	1.713
Quadratic	median income	0.219	1.697
Linear	teachers	0.030	not computed
Multiple (five)	unemployment, college, married, income, lunch	0.628	1.145
Multiple (six)	the five above + teachers	0.632	1.136
Reduced	unemployment, college, lunch	0.628	1.146

Keeping unemployment, college, and lunch share gives all significant coefficients and essentially the same fit as the five-predictor model. The nested F-test finds no significant loss ($F = 0.215$, $p = 0.807$; $\Delta SSR = 1.0$ on 2 degrees of freedom).

Table 3. Coefficients

Term (reduced model)	Standardized coefficient	p-value
Intercept	20.299	< 0.001
rate_unemployment	−0.123	< 0.001
percent_college	0.283	< 0.001
percent_lunch	−1.777	< 0.001

The percent_lunch predictor variable is the best predictor of ACT score, with a standarized coefficient of −1.777.

The relationship between percent_lunch and charter school was compared, although the data included mostly non-charter schools. Charter schools had a higher median percent lunch (about 0.6) than non-charter schools (about 0.37). Compared with non-charter schools, charter schools had an average ACT score 0.21 points lower, which is not significant.

The relationship between percent_lunch and number of teachers was also examined. The regression lines of average ACT score on percent lunch are close to parallel across the four teacher groups, and this relationship also did not appear to be significant.

Discussion

The main result of this analysis is that ACT scores can be predicted by socioeconomic factors. The strongest factor included in this analysis was the percent of students receiving free or reduced lunch. An increase of one standard deviation in percent_lunch lowers the predicted ACT score by 1.78 points. Median income is a significant predictor alone but it is not significant once the other variables are included. All of the factors from the EdGap data probably measure overlapping aspects of community economic disadvantage, and percent lunch captures it most directly in this instance. Number of teachers per school was included to examine other potential bias that might be captured by schools size — for example, small rural schools or small private schools. However, teachers in the six-predictor model raised R^2 by 0.004, and this does not seem to be a large factor on the scale of percent of students receiving free or reduced lunch.

This analysis has several limitations. A limited set of data was used, and the statistical models were not verified on any new data, so it is unclear how accurately the models would predict ACT scores for new schools. Missing socioeconomic values were imputed with a model fit on the existing dataset. The data covers only a subset of U.S. high schools, and the high schools it does include are from certain areas. So, it is unclear if the model is accurate for other U.S. schools.

Also, the predictor variables here are only roughly related to larger socioeconomic factors. Percent lunch is a rough approximation for economic hardship. Other factors like rurality are approximated by number of teacher / school size, but this is a poor approximation, and we can not be sure how well we are actually capturing the socioeconomic factors.

For further analysis, we could add more data to our initial data set. The model could be tested on new data, or data from different school years. More categorical variables could also be examined to see their effect on the data.

Conclusion

This project found that socioeconomic factors predict the average ACT score of a school fairly well. A linear model with three socioeconomic predictors explained 62.8% of the variance in average ACT score with a mean absolute error of about 1.15 points. The percent of students receiving free or reduced price lunch was the strongest predictor.

References

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